Sample Job Description

Position: AI / Machine Learning Engineer

Department:	Artificial Intelligence & Machine Learning	**Job Type:**	Full-Time / Campus Graduate
Location:	Bangalore / Bhubaneswar (Hybrid)	**Experience:**	0 - 2 Years (Freshers Welcome)

Job Overview & Company Mission

We are looking for an ambitious and innovative **AI Engineer** to join our Core Intelligence Engineering team. Our team develops cutting-edge Generative AI platforms, Retrieval-Augmented Generation (RAG) frameworks, and production-grade Natural Language Processing (NLP) solutions. In this role, you will architect intelligent pipelines, collaborate with backend teams to serve low-latency AI endpoints, and design knowledge-graph augmented machine learning systems.

Specific Responsibilities

- Design, develop, and fine-tune Machine Learning and Deep Learning models using **Python** and **PyTorch**.
- Implement production Retrieval-Augmented Generation (**RAG**) pipelines utilizing vector databases and semantic embedding models.
- Build scalable, high-throughput backend microservices and RESTful APIs using **FastAPI** and Uvicorn.
- Containerize AI services with **Docker** and deploy reliable workloads across cloud infrastructure and **Kubernetes** clusters.
- Set up automated **CI/CD pipelines** using **GitHub Actions** for automated unit testing, linting, and continuous model delivery.
- Design relational database schemas and manage transactional persistence using **PostgreSQL** and SQL query optimization.
- Optimize inference latency, memory consumption, and throughput for deep learning models running on GPU and CPU accelerators.
- Collaborate cross-functionally with frontend teams and product managers to integrate machine intelligence seamlessly into user interfaces.

Core Technical Competencies & Skills

- **Programming & Foundations:** Strong proficiency in Python, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA).
- **Machine Learning & Generative AI:** PyTorch, Scikit-Learn, Transformers (Hugging Face), LLM fine-tuning, Prompt Engineering, RAG.
- **API & Backend Engineering:** FastAPI, REST APIs, Microservices Architecture, Asyncio, JSON schema validation.
- **Cloud & DevOps:** Docker, Containerization, Kubernetes, Cloud Infrastructure (AWS/GCP), GitHub Actions, CI/CD Pipelines.
- **Data & Databases:** PostgreSQL, Relational Databases, SQL, Vector Stores (FAISS/ChromaDB), Data Preprocessing, PyPDF.

Qualifications & Requirements

- Bachelor's or Master's degree in Computer Science, Artificial Intelligence, Data Science, or a related engineering discipline.
- Demonstrated project portfolio in Machine Learning, Generative AI, or Natural Language Processing (e.g., GitHub repositories or live apps).
- Familiarity with version control workflows using Git and collaborative development.
- Strong problem-solving capability, self-starter mentality, and passion for emerging AI technologies.
- Excellent written and verbal communication skills to explain complex architectural decisions clearly.